

# CS-3009 Computer Architecture-Spring 2026

## Assignment#1

Due Date and Time: 11<sup>th</sup> February, 2026 (08:30 AM)

Weight: 3.33%

### Instructions:

1. *Late assignment will not be accepted*
2. *Only handwritten attempt will be graded, i.e., printed attempts will not be graded*
3. *There will be no credit if the given requirements are changed*
4. *Your solution will be evaluated in comparison with the best solution (having minimum number of steps) for part 3*
5. *Please mention the question number and its part (if any) before writing down its solution. Use the same conventions used in the assignment's document.*
6. *Don't plagiarize. Plagiarism would lead to straight zero in the assignment regardless of the percentage copied.*
7. *Also submit scanned copies of your assignment on GCR. **20% penalty for not submitting it on GCR***
8. *Your evaluation is based on the quiz from this assignment. So be ready for quiz on the submission day.*

### Question # 1

Consider four different processors P1, P2, P3 and P4 executing the same instruction set. P1 has a 2 GHz clock rate and a CPI of 1.5. P2 has a 2.5 GHz clock rate and a CPI of 1.0. P3 has a 2.5 GHz clock rate and has a CPI of 2.2. P4 has a 4.0 GHz clock rate and has a CPI of 2.0.

- a. Which processor has the highest performance expressed in instructions per second?

- b. If the processors each execute a program in 15 seconds, find the number of cycles and the number of instructions.

- c. We are trying to reduce the execution time by 40%. Discuss is it possible to improve execution time by 40% Does it affect CPI? What clock rate should we have to get this time reduction?

### **Question # 2**

Consider two different implementations, M1 and M2, of the same instruction set. There are three classes of instructions (A, B, and C) in the instruction set. M1 has a clock rate of 1500 MHz and M2 has a clock rate of 2000 MHz. The average number of cycles for each instruction class and their frequencies (for a typical program) are as follows:

| Instruction Class | Machine M1-<br>Cycles/Instruction | Machine M2-<br>Cycles/Instruction | Frequency |
|-------------------|-----------------------------------|-----------------------------------|-----------|
| A                 | 6                                 | 1                                 | 50%       |
| B                 | 5                                 | 3                                 | 30%       |
| C                 | 2                                 | 4                                 | 20%       |

- a. What is the average CPI for each machine, M1, and M2?

- b. Calculate the average MIPS ratings for each machine, M1 and M2.

- c. Which machine has a smaller MIPS rating? Which individual instruction class CPI do you need to change, and by how much, to have this machine have the same or better performance

as the machine with the higher MIPS rating (you can only change the CPI for one of the instruction classes on the slower machine)?

### **Question # 3**

Compilers can have a profound impact on the performance of an application. Assume that for a program, compiler A results in a dynamic instruction count of  $3.2E9$  and has an execution time of 1.8 s, while compiler B results in a dynamic instruction count of  $2.1E9$  and an execution time of 2.5 s.

- a. Find the average CPI for each program given that the processor has a clock cycle time of 0.5 ns.

- b. Assume the compiled programs run on two different processors. If the execution times on the two processors are the same, how much faster is the clock of the processor running compiler A's code versus the clock of the processor running compiler B's code?

- c. A new compiler is developed that uses only  $1.5E8$  instructions and has an average CPI of 1.1. What is the speedup of using this new compiler versus using compiler A or B on the original processor?

#### **Question # 4**

Assume for arithmetic, load/store, and branch instructions, a processor has CPIs of 1, 12, and 5, respectively. Also assume that on a single processor a program requires the execution of  $2.56\text{E}9$  arithmetic instructions,  $1.28\text{E}9$  load/store instructions, and 256 million branch instructions.

Assume that each processor has a 2 GHz clock frequency.

Assume that, as the program is parallelized to run over multiple cores, the number of arithmetic and load/store instructions per processor is divided by  $0.7 \times p$  (where  $p$  is the number of processors) but the number of branch instructions per processor remains the same.

- a. Find the total execution time for this program on 1, 2, 4, and 8 processors, and show the relative speedup of the 2, 4, and 8 processor result relative to the single processor result.

- b. If the CPI of the arithmetic instructions was doubled, what would the impact be on the execution time of the program on 1, 2, 4, or 8 processors?

- c. To what should the CPI of load/store instructions be reduced in order for a single processor to match the performance of four processors using the original CPI values?

### **Question # 5**

The results of the SPEC CPU2006 bzip2 benchmark running on an AMD Barcelona has an instruction count of  $2.389 \times 10^{12}$ , an execution time of 750 s, and a reference time of 9650 s.

**a.** Find the CPI if the clock cycle time is 0.333 ns.

**b.** Find the increase in CPU time if the number of instructions of the benchmark is increased by 10% without affecting the CPI.

**c.** Find the increase in CPU time if the number of instructions of the benchmark is increased by 10% and the CPI is increased by 5%.

**d.** Suppose that we are developing a new version of the AMD Barcelona processor with a 4 GHz clock rate. We have added some additional instructions to the instruction set in such a way that the number of instructions has been reduced by 15%. The execution time is reduced to 700 s. Find the new CPI.

- e. This CPI value is larger than obtained in part (a) as the clock rate was increased from 3 GHz to 4 GHz. Determine whether the increase in the CPI is similar to that of the clock rate. If they are dissimilar, why?

- f. By how much has the CPU time been reduced?

- g. For a second benchmark, libquantum, assume an execution time of 960 ns, CPI of 1.61, and clock rate of 3 GHz. If the execution time is reduced by an additional 10% without affecting to the CPI and with a clock rate of 4 GHz, determine the number of instructions.

- h. Determine the clock rate required to give a further 10% reduction in CPU time while maintaining the number of instructions and with the CPI unchanged.

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- i. Determine the clock rate if the CPI is reduced by 15% and the CPU time by 20% while the number of instructions is unchanged.

### **Question # 6**

Consider the following two processors. P1 has a clock rate of 4 GHz, average CPI of 0.9, and requires the execution of  $5.0E9$  instructions. P2 has a clock rate of 3 GHz, an average CPI of 0.75, and requires the execution of  $1.0E9$  instructions.

- a. One usual fallacy is to consider the computer with the largest clock rate as having the largest performance. Check if this is true for P1 and P2.

- b. Another fallacy is to consider that the processor executing the largest number of instructions will need a larger CPU time. Considering that processor P1 is executing a sequence of  $1.0E9$  instructions and that the CPI of processors P1 and P2 do not change, determine the number of instructions that P2 can execute in the same time that P1 needs to execute  $1.0E9$  instructions.

- c. A common fallacy is to use MIPS (millions of instructions per second) to compare the performance of two different processors, and consider that the processor with the largest MIPS has the largest performance. Check if this is true for P1 and P2.

- d. Another common performance figure is MFLOPS (millions of floating-point operations per second), defined as
- $$\text{MFLOPS} = \text{No. FP operations} / (\text{execution time} \times 1\text{E}6)$$
- but this figure has the same problems as MIPS. Assume that 40% of the instructions executed on both P1 and P2 are floating-point instructions. Find the MFLOPS figures for the programs.

### **Question # 7**

Another pitfall is expecting to improve the overall performance of a computer by improving only one aspect of the computer. Consider a computer running a program that requires 280 s, with 90 s spent executing FP instructions, 120 s executed L/S instructions, and 70 s spent executing branch instructions.

- a. By how much is the total time reduced if the time for FP operations is reduced by 20%?

- b. By how much is the time for INT operations reduced if the total time is reduced by 20%?



c. Can the total time can be reduced by 20% by reducing only the time for branch instructions?

### **Question # 8**

Assume a program requires the execution of  $40 \times 120$  FP instructions,  $50 \times 120$  INT instructions,  $90 \times 120$  L/S instructions, and  $15 \times 120$  branch instructions. The CPI for each type of instruction is 1, 1, 4, and 2, respectively.

Assume that the processor has a 2.5 GHz clock rate.

a. By how much must we improve the CPI of FP instructions if we want the program to run two times faster?

b. By how much must we improve the CPI of L/S instructions if we want the program to run two times faster?

- c. By how much is the execution time of the program improved if the CPI of INT and FP instructions is reduced by 40% and the CPI of L/S and Branch is reduced by 30%?

### **Question # 9**

(Amdahl's law question) Suppose you have a machine which executes a program consisting of 45% floating point multiply, 30% floating point divide, and the remaining 25% are from other instructions.

- a. Management wants the machine to run 2 times faster. You can make the divide run at most 3 times faster and the multiply run at most 5 times faster. Can you meet management's goal by making only one improvement, and which one?

- b. Dogbert has now taken over the company removing all the previous managers. If you make both the multiply and divide improvements, what is the speed of the improved machine relative to the original machine?

- c. Suppose that we can improve the floating point instruction performance of machine by a factor of 3 (the same floating point instructions run 3 times faster on this new machine). What percent of the instructions must be floating point to achieve a Speedup of at least 4?

### **Question # 10**

#### **A two-part question:**

##### **(Part A)**

Assume that a design team is considering enhancing a machine by adding MMX (multimedia extension instruction) hardware to a processor. When a computation is run in MMX mode on the MMX hardware, it is 5 times faster than the normal mode of execution. Call the percentage of time that could be spent using the MMX mode the percentage of media enhancement.

- a. What percentage of media enhancement is needed to achieve an overall speedup of 2?

- b. What percentage of the run-time is spent in MMX mode if a speedup of 2 is achieved? (Hint: You will need to calculate the new overall time.)

- c. What percentage of the media enhancement is needed to achieve one-half the maximum speedup attainable from using the MMX mode?

**(Part B)**

If processor A has a higher clock rate than processor B, and processor A also has a higher MIPS rating than processor B, explain whether processor A will always execute faster than processor B. Suppose that there are two implementations of the same instruction set architecture. Machine A has a clock cycle time of 25ns and an effective CPI of 2.5 for some program, and machine B has a clock cycle time of 10ns and an effective CPI of 1.5 for the same program. Which machine is faster for this program, and by how much?